



Tools in geometry

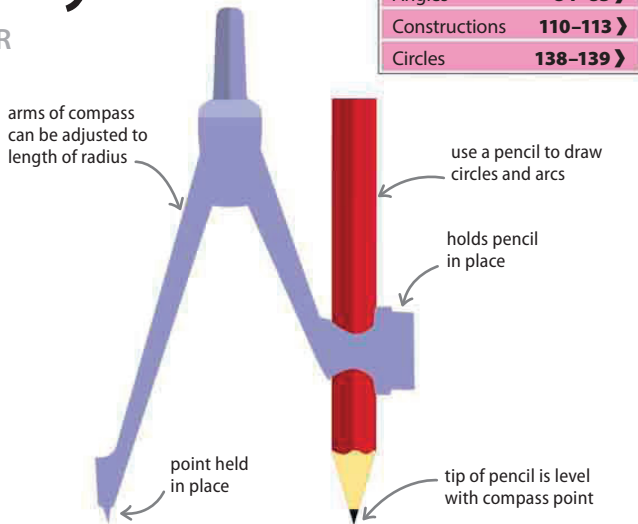
MATHEMATICAL INSTRUMENTS ARE NEEDED FOR MEASURING AND DRAWING IN GEOMETRY.

Tools used in geometry

Tools are vital to measure and construct geometric shapes accurately. The essential tools are a ruler, a compass, and a protractor. A ruler is used for measuring and to draw straight lines. A compass is used to draw a whole circle or a part of a circle (called an arc). A protractor is used to measure and draw angles.

Using a compass

A tool for drawing circles and arcs, a compass is made up of two arms attached at one end. To use a compass, hold the arm that ends in a point still, while pivoting the other arm, which holds a pencil, around it. The point becomes the center of the circle.

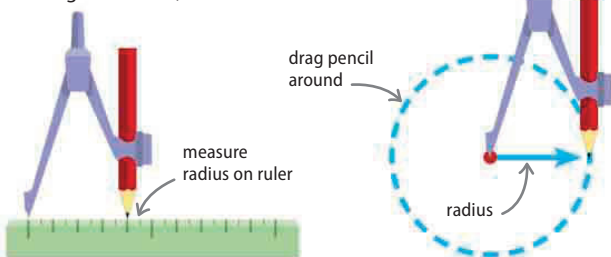


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▽ Drawing a circle when given the radius

Set the distance between the arms of a compass to the given radius, then draw the circle.

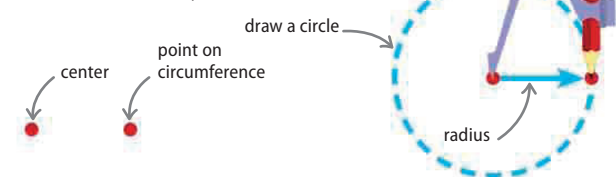


Use a ruler to set the arms of the compass to the given radius.

With the compass set to the radius, hold the point down and drag the pencil around.

▽ Drawing a circle when given its center and one point on the circumference

Put the point of the compass where the center is marked and extend the other arm so that the tip of the pencil touches the point on the circumference. Then, draw the circle.

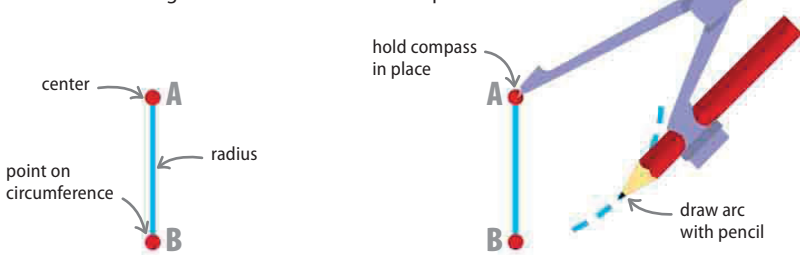


Set the compass to the distance between the two points.

Hold the point of the compass down and draw a circle through the point.

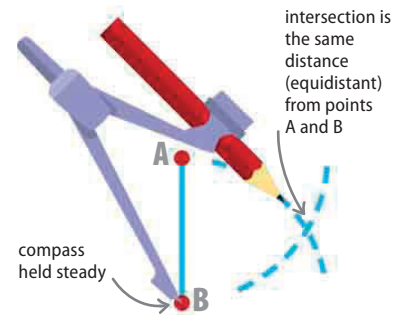
▽ Drawing arcs

Sometimes only a part of a circle—an arc—is required. Arcs are often used as guides to construct other shapes.



Draw a line and mark the ends with a point—one will be the center of the arc, the other a point on its circumference.

Set the compass to the length of the line—the radius of the arc—and hold it on one of the points to draw the first arc.



Draw a second arc by holding the point of the compass on the other point. The intersection is equidistant from A and B.